Mansi Sheth

Security Researcher · Applied Cryptography

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PROFILE

Security researcher and developer with eighteen years across application security, applied cryptography, vulnerability intelligence, and static analysis. One habit runs through all of it: find the hard problem, go deep, and share what you learn. **Three granted patents** and **10+ peer-reviewed conference talks** reflect that. Currently focused on **intersection of AI X cybersecurity**, both as a new toolset for long-standing hard problems and as a source of new ones. Open to research and development roles where deep security expertise and AI come together.

HIGHLIGHTS

- **3 Granted Patents** — Static analysis, compiler tooling, software security
- **10+ Conference Talks** — DEF CON · NorthSec · SecTor · JavaOne · AppSec USA · JavaZone · XML Prague
- **Java Crypto Blog Series** — Practical cryptography implementation guides (formerly on Veracode Engineering Blog)

  → *Full details at 1mansis.github.io*

PROFESSIONAL EXPERIENCE

Member of Technical Staff · VulnCheck *2025 – Present*

- Build the data backbone that customers' **AI pipelines** run on. Research and maintain ingestion across **600+ vulnerability intelligence sources**, and own the normalization and quality standards that keep the output reliable for AI triage and downstream tooling.
- Built an AI harness pipeline that catches silent security fixes in open-source commits, with a second harness verifying each classification.
- Shaped VulnCheck's **AI roadmap** by defining use cases and architecture for the official VulnCheck MCP Server; and sole author of VulnCheck Agent Tools, Claude Code skills for vulnerability intelligence in engineering workflows.
- Built and deployed an **ML classifier (TF-IDF + Logistic Regression)** to categorize exploit types within VulnCheck's exploit taxonomy, now live on customer endpoints; used **LLM-assisted labeling** to expand training coverage for sparse categories.

Lead Cryptography Software Engineer · SandboxAQ *2023 – 2024*

- Technical Lead for Sandwich: a **crypto-agile, post-quantum-ready** library built to cover all cryptographic primitives — asymmetric encryption, digital signatures, hashing — with swappable classical and PQC backends across C/C++, Python, Go, and Rust; the TLS layer was open-sourced before the project concluded (github.com/sandbox-quantum/sandwich).
- Architected the swappable crypto-backend pattern enabling seamless algorithm transitions — classical to PQC — across C/C++, Python, Go, and Rust bindings; validated the design by integrating Sandwich into internal SandboxAQ products to achieve crypto-agility in practice.
- Led a team of **4 cryptography engineers** — directing technical work, resolving implementation blockers, and keeping the team aligned to product requirements and enterprise adoption goals.

Sr. Principal Security Researcher · Veracode *2013 – 2023*

Security Research

- Primary researcher for the **Java ecosystem** — Veracode's highest-revenue platform — owning security research and detection coverage across the full **CWE and OWASP** vulnerability taxonomy, spanning the JDK, major application and web frameworks, and the vast open-source library landscape that enterprise Java applications depend on.

- Pioneered security research for Python and Go at Veracode — the first to establish vulnerability detection coverage for each language, going deep on ecosystem-specific security anti-patterns, CWE mappings, and detection strategies across their respective framework and library landscapes; developed supporting research tooling to enable scalable analysis.
- Distilled deep Java cryptography research into a language-agnostic blueprint — defining detection logic, flaw categorization, and remediation guidance for crypto vulnerabilities — adopted as the standard specification across all languages and frameworks.
- Formally designated by management as technical mentor for new hires and interns — responsible for onboarding, task assignment, and technical direction within the research team.

Research & Development

- Received **Thought Leadership award** for the highest number of approved patents in company history; patented **3 novel compiler and tooling approaches** for static analysis of non-grammar-based and templating languages.
- Architected a secure research data pipeline — EC2-hosted binary metadata extraction, parsed and stored in a Native XML Database for structured querying, with client-side cryptographic protections and KMS-managed keys for encrypted storage back to S3.
- Leveraged the pipeline for **prevalence analysis** — identifying what frameworks, libraries, and APIs were actually in use across the customer base, allowing research to focus on the highest-impact areas of large ecosystems.
- Applied **ML + NLP** to locate security-sensitive functionality within codebases — enabling research to treat vulnerabilities discovered in those areas with higher investigation priority.

Sr. Technology Analyst, Enterprise Application Security · Fidelity Investments *2007 – 2013*

Penetration Testing

- Conducted security assessments and penetration testing on Fidelity's critical flagship web applications, APIs, and internal infrastructure; led targeted red-team exercises across high-value financial systems.
- Administered and extended AppScan and Burp Suite for enterprise pen testing — writing Burp plugins for cross-tool result aggregation and reporting, and exploring JavaScript automation for session handling within AppScan workflows.

Static Analysis & Tooling

- Lead developer of **'Assembly Line'**: an internal SAST orchestration platform extending commercial static analysis tools with custom Fortify rules, bridging coverage gaps across Fidelity's proprietary frameworks.
- Developed deep expertise in Fidelity's custom application frameworks and Fortify's detection engine — writing custom rules to automate vulnerability discovery across **200+ applications** built on those frameworks.
- Leveraged that framework knowledge to advise security architecture teams on anti-patterns and secure design recommendations — directly influencing how Fidelity's internal frameworks were hardened.

Open Source

- Contributor to OWASP iGoat; authored the Keychain exercise module used in mobile security training.

Lead Software Developer · Intralinks · Accenture · IRIS Ltd *2002 – 2007*

- At Accenture, worked exclusively on JP Morgan's Futures and Options modernization project — and led security remediation of a comprehensive third-party penetration test report, owning end-to-end fixes across a broad range of application security vulnerabilities with a team of two junior engineers. First hands-on introduction to application security.
- At IRIS Ltd, developed dynamic web products for mid-size Indian businesses and contributed a core module to HDFC Bank's platform.

M.S. Computer Science · University of Massachusetts Lowell *2011*

B.E. Computer Engineering · University of Mumbai *2002*